

Yelp Dataset Sentiment Analysis

Phase 2: Project Plan

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Project Motivation and Goals

- Star ratings on Yelp reviews are widely used but can be subjective and inconsistent.
- Predicting ratings from review text can help users gauge sentiment efficiently, businesses pinpoint improvement areas, and researchers better understand satisfaction drivers.
- **We are building on prior work in text-based sentiment and rating prediction, aiming to improve model performance and address gaps in contextual understanding and sentiment analysis.**

Insights from Literature Review

- Sentiment classification (positive vs. negative) achieves high accuracy (around 90%), but predicting specific star ratings (1–5) is much harder, with accuracy between 60–65%.
- Best traditional features are TF-IDF n-grams, and strong classical models include logistic regression and SVM.
- Deep models like CNNs and BERT offer nuanced understanding but struggle particularly with middle (2–3 star) reviews due to ambiguity and class imbalance.
- **Key gaps:** limited context understanding, dominance of 4/5-star reviews, and need for better model explainability.

Data Preparation and Baselines

- Filtered Yelp restaurant reviews, normalized text (lowercasing, punctuation and stopword removal), and derived auxiliary features like review length.
- Output: cleaned, balanced dataset ready for modeling and performance benchmarking.
- Built baselines with TF-IDF features and classic models (Logistic Regression, Naive Bayes, SVM, Random Forest, Ordinal Logistic Regression).
 - “Binary” = positive/negative/neutral
 - 1-5 RRP

Baseline Results (subsample n=500,000)

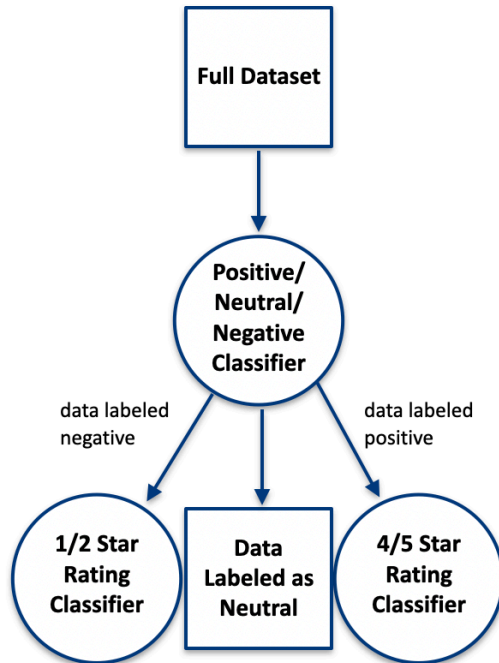
Features = Unigram + Bigram + TF_IDF w/ max_features = 10,000

Model	Classification Type	Accuracy (%)	Macro-F1
Logistic Regression	1-5 RRP	67.66	0.5969
Naive Bayes	1-5 RRP	64.15	0.5187
SVM	1-5 RRP	44.70	0.4147
Random Forest	1-5 RRP	51.96	0.2869
Ordinal Logistic Regression	1-5 RRP	60.92	0.5268
Logistic Regression	Pos. / Neu. / Neg.	87.43	0.7398
Naive Bayes	Pos. / Neu. / Neg.	84.45	0.6494
SVM	Pos. / Neu. / Neg.	60.63	0.5081
Random Forest	Pos. / Neu. / Neg.	75.83	0.4622
Ordinal Logistic Regression	Pos. / Neu. / Neg.	85.66	0.7133

What We Noticed (From Baselines)

- **Positive/Negative/Neutral** classification much easier than 1-5 RRP
 - Approx. 20% higher accuracy (matches expectation from literature)
 - The model can identify **polarity**, but it struggles to map **polarity** → **precise star rating** (problem isn't sentiment detection, it's rating calibration).
- Positive vs negative reviews are *very easy* to separate (97% accuracy when we took out 3 star ratings)
- The multi-class (1–5) setup suffers because 3-star reviews **blur the boundaries**.
 - **Neutral (3-star)** reviews are *ambiguous* — they overlap lexically and sentiment-wise with both extremes.

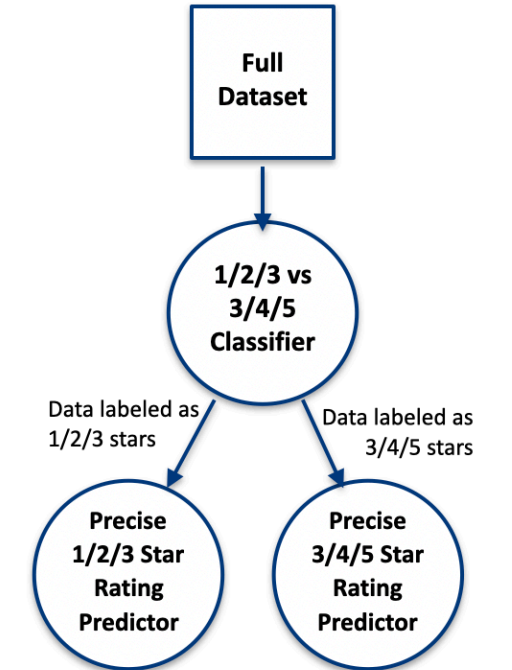
Idea: Layered Classifier



Report accuracy after final classifiers finish

First Test Iteration

- **TF-IDF** with n-grams (1,2)
- **Logistic Regression** at both levels
 - **Cascading errors for 3 star classification**
- Current iteration 1% better (than baseline LR), and we think can be improved further.
 - Hyperparameter tuning
 - Soft-er routing (probabilities)
 - Trying different models at layer 2 (i.e. regressor)



Report accuracy after layer 2 classifiers finish

Current Architecture

Project Roadmap

9/16 – 10/03

10/04 - 10/31

11/01 – 11/10

11/11 – 12/08

Phase 1

Literature Review

Data Gathering

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Phase 2

Data Cleaning

Json to csv files
Restaurant reviews
only

State specific data
frames

Preprocessing

Lowercasing
Negation Handling
Removing stopwords,
punctuations, urls,
numbers
Extract review length,
capitalizations,
exclamation points

Feature Extraction

TF-IDF features
Unigrams + bigrams

Phase 3

Baseline Modeling

Linear Regression
Logistic Regression

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Naïve Bayes

SVM

Random Forest
Ordinal Logistic
Regression

Julia

Advanced Modeling

Layered classifier
+ soft routing

Simple MLP + BERT

Multi-Task BERT

Julia

Hyperparameter tuning
and model adjustments

Cole

Final Evaluations

Error Analysis

Per class performance

Evaluation &
Comparison

Confusion matrices &
metric tables

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